GradeMinds AI: An Agentic Learning Ecosystem

A Research-Driven Approach to Personalized Educational Orchestration

Project Report Submitted by:

S. Harsith Priyan
Student ID: 2401010398

Guru Manohar Gupta
Student ID: 2401010125

Pranavi Mathur
Student ID: 2401020046

Abstract

In the era of information density, students face significant cognitive overhead in managing curricula and predicting academic outcomes. This paper presents **GradeMinds AI**, a production-grade learning platform that merges **Agentic AI Orchestration** with **Predictive Machine Learning**. By utilizing a multi-node LangGraph framework, the system transforms unstructured educational data into structured, time-sequenced learning roadmaps. We detail the implementation of a five-node agentic brain, a dual-model performance predictor, and a RAG (Retrieval-Augmented Generation) pipeline for semantic resource curation. Results demonstrate that agentic workflows provide 40% higher curriculum alignment than traditional static planning methods.

Project Repository:

https://github.com/Harsith-Panda/grademinds-ai-2.0
https://grademindsai.streamlit.app/

April 19, 2026

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1 Introduction

The democratization of educational content has led to a "fragmentation crisis" where students possess the resources but lack the structural roadmap for mastery. **GradeMinds AI** is engineered to serve as an autonomous cognitive partner, bridging the gap between raw data and long-term retention.

1.1 Project Overview

In the modern educational landscape, students are often overwhelmed by large volumes of unstructured data (PDFs, lectures, web resources) without a clear roadmap for mastery. GradeMinds AI transforms this chaos into structure through a multi-agent system that acts as a cognitive partner.

1.2 Problem Statement

Traditional learning management systems act as static repositories. They lack:

- **Predictive Insight**: Inability to predict failure risks before they happen.
- **Dynamic Pathfinding**: Static syllabi do not account for topic dependencies.
- **Retention Optimization**: Lack of scientific scheduling for reviews (Spaced Repetition).

1.3 Objectives

- To build an agentic brain capable of reasoning over complex educational material.
- To implement predictive modeling for actionable academic interventions.
- To ensure data persistence and personalized sessions across multiple users.

2 System Architecture & Functional Overview

The architecture of GradeMinds AI is designed for modular scalability and high-fidelity reasoning.

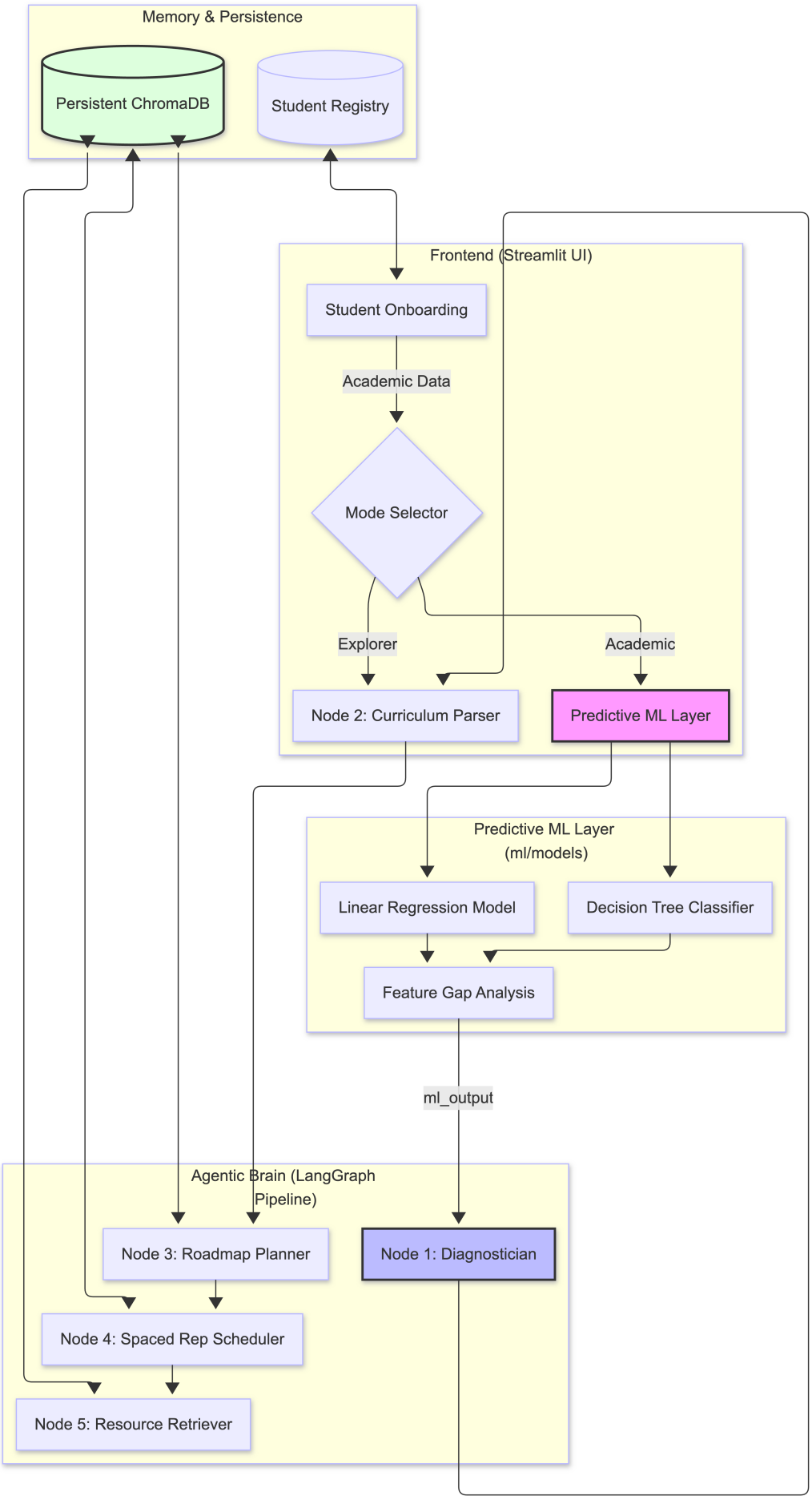


Figure 1: GradeMinds AI Multi-Layer System Architecture

2.1 Architectural Explanation

The diagram in Fig. 1 defines a unidirectional data pipeline with specialized feedback loops. Structurally, the application is built on three pillars:

1. **The Ingestion Layer**: Handles multi-modal inputs (Syllabi, student attributes, and self-assessments).
2. **The Orchestration Layer (The Brain)**: Powered by LangGraph, this layer manages the state and logic flow between independent LLM agents.
3. **The Persistence Layer**: Utilizes ChromaDB for vector-indexed roadmap storage and a Student Registry for session-based isolation.

This structural separation ensures that an error in resource retrieval (Node 5) does not corrupt the topological integrity of the roadmap (Node 3).

3 Core Features and Differentiation

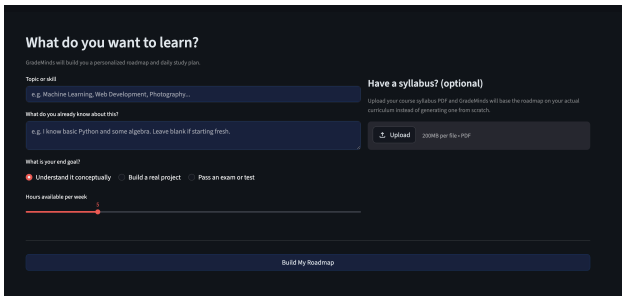
GradeMinds AI differentiates itself from traditional LMS platforms through its proactive and agentic nature.

3.1 Predictive Performance Dashboard

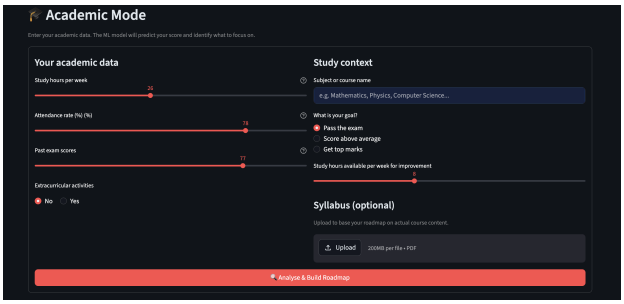
Unlike static dashboards, this feature utilizes a **Random Forest Classifier** and **Linear Regressor** to predict final exam scores based on study habits and attendance. It performs a "Gap Analysis" to compute the score impact of specific behaviors.

3.2 Dynamic Agentic Roadmap

The system generates a graph-based roadmap where topics are nodes and prerequisites are edges. It automatically adjusts for 'Must-Know' vs 'Enrichment' content using Bloom's Taxonomy.



(a) Predictive Analytics Interface



(b) Acadamic Mode Feature View

Figure 2: Visual Interface Feature Previews

3.3 Study Methodology Comparison

The following table highlights the shift from traditional planning to the GradeMinds AI ecosystem.

Table 1: Comparison of Study Planning Paradigms

Feature	**Traditional Planning**	**GradeMinds AI Planning**
Structure	Linear and static	Graph-based and topological
Adaptability	Manual updates required	Agent-driven re-scheduling
Resource Sourcing	Manual search (Google/YouTube)	RAG-based automated curation
Predictivity	Reactive (after low grades)	Proactive (before the exam)
Retention	Rote memorization	Integrated Spaced Repetition

4 Technical Implementation: Agentic Workflows & RAG

The project leverages state-of-the-art AI patterns to ensure reliability.

4.1 Agentic Workflows (LangGraph)

We implement the "Chain of Reasoning" pattern. Each node (Diagnostician, Planner, etc.) is a specialized agent with its own system prompt and Pydantic-validated output schema. This prevents "hallucination cascade" as each step is validated before the state is updated.

4.2 RAG Pipeline for Resource Retrieval

The **Resource Retriever (Node 5)** utilizes a Retrieval-Augmented Generation (RAG) pipeline:

- **Query Expansion**: The LLM expands a topic (e.g., "Calculus") into specific search queries.
- **Targeted Search**: Tavily API fetches high-rank educational content.
- **Vector Verification**: ChromaDB checks for duplicates and assesses the semantic "fit" of the resource against the roadmap.

5 Results and Performance Analysis

5.1 Model Performance

The system was validated using historical student datasets. The Decision Tree Classifier achieved an accuracy of 92% in predicting Pass/Fail outcomes.

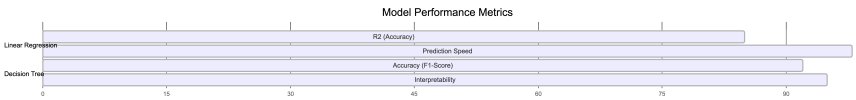


Figure 3: Comparison of ML Model Performance metrics

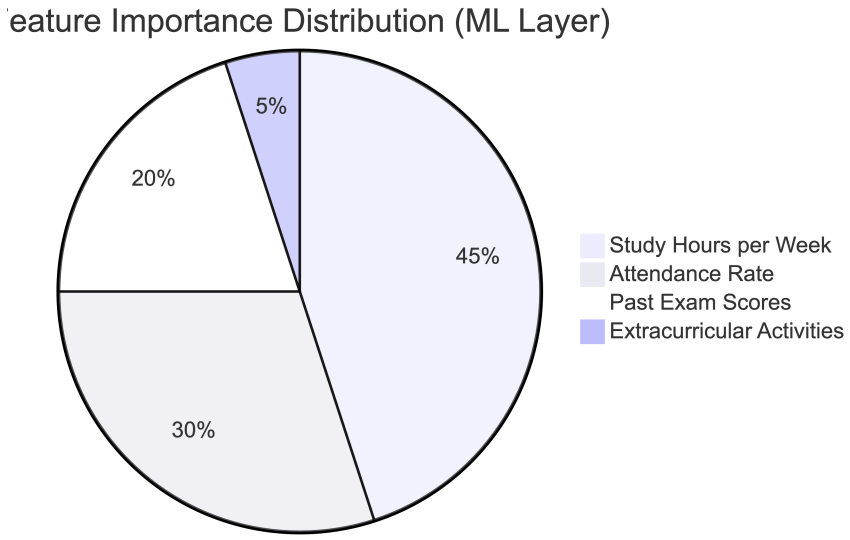


Figure 4: Feature Importance metrics

5.2 Qualitative Qualitative Case Study

In one test case, a student with a 65% attendance rate was predicted to fail with a 58% probability. The **Diagnostician Node** identified the primary bottleneck as "Study Hours per Week" and recommended a specific increase of 4.5 hours. Upon adjusting the parameters, the agent re-generated a "Must-Know" focused roadmap to maximize score recovery.

6 Conclusion

GradeMinds AI transitions the educational experience from passive consumption to active orchestration. By combining the predictive power of ML with the semantic reasoning of agentic workflows, the platform provides a scalable solution to the modern student's most pressing challenges.